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MIL-L-49320(CR)
14 April 1993

# MILITARY SPECIFICATION

## LASER INFRARED OBSERVATION SET, AN/PVS-6

**This specification is approved for use by the US Army Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.**

## 1. SCOPE

1.1 Scope. This specification covers the Laser Infrared Observation Set, referred to herein as the Miniature Eyesafe Laser Infrared Observation Set (MELIOS).

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation, (see 6.2.h).

SPECIFICATIONS

MILITARY

| | | |
|---|---|---|
| MIL-P-11268 | - | Parts, Materials, and Processes Used in Electronic Equipment |
| MIL-M-13231 | - | Marking of Electronic Items |
| MIL-C-46168 | - | Coating, Aliphatic Polyurethane, Chemical Agent Resistant |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-ED-TM, Fort Monmouth, NJ 07703-5023 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A: FSC 5860

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| | | |
|---|---|---|
| MIL-B-49436/6 | - | Battery, Rechargeable, Sealed Nickel Nickel Cadmium, BB-516/U |
| MIL-B-49461/1 | - | Battery Non-Rechargeable, Lithium Thionyl Chloride, BA-6516/U |

STANDARDS

MILITARY

| | | |
|---|---|---|
| MIL-STD-454 | - | Standard General Requirements for Electronic Equipment |
| MIL-STD-461 | - | Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference |
| MIL-STD-462 | - | Electromagnetic Interference Characteristics, Measurement of |
| MIL-STD-781 | - | Reliability Testing for Engineering Development Qualification and Production |
| MIL-STD-810 | - | Environmental Test Methods and Engineering Guidelines |
| MIL-STD-882 | - | System Safety Program Requirements |
| MIL-STD-2000 | - | Standard Requirements for Soldered Electrical and Electronic Assemblies |

HANDBOOK

| | | |
|---|---|---|
| MIL-HDBK-781 | - | Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production |

(Unless otherwise indicated, copies of federal and military specification, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Bldg 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation, (see 6.2.h).

PUBLICATIONS

| | | |
|---|---|---|
| Title 10 | - | United States Code of Federal Regulations Energy |
| TB-MED-524 | - | Control of Hazards to Health from Laser Radiation |



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| | | |
|---|---|---|
| TM 11-5860-202-10 | - | Operators Manual, Mini Eyesafe Laser Infrared Observation Set (MELIOS), AN/PVS-6 |

DRAWINGS

| | | |
|---|---|---|
| A3187100 | - | Mini Eyesafe Laser Infrared Observation Set |
| A3187116 | - | Reticle |
| A3187120 | - | Laser Rangefinder, MELIOS (AN/PVS-6) |
| A3187229 | - | Case, Laser Rangefinder |
| A3187238 | - | Tripod, Assembly |
| A3187239 | - | Shipping and Storage Container Assembly MELIOS (AN/PVS-6) |

(Copies of other Government documents, drawings, and publications required by suppliers in performance of specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document shall take precedence. Nothing in this document, however supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The MELIOS Laser Rangefinder herein identified as the MLRF is a hand held, self contained, laser ranging system. The MLRF consists of the Laser Rangefinder, MELIOS (A3187120) less all ancillary equipment listed in 3.7.13.

3.2 Qualification. Unless otherwise specified by the contract or purchase order MELIOS units furnished under this specification shall be products approved as qualified products (see 3.12.6 and 6.4).

3.2.1 First article. When specified (see 6.2.b), units shall be subjected to Initial Production Testing (IPT), (see 4.4).

3.3 Parts, materials and processes. Parts, materials and processes shall be as specified herein and as shown on drawing A3187100 and other applicable drawings. Material not specified shall be selected by the contractor and shall be subjected to all provisions of this specification and shall conform to MIL-P-11268.

3.4 Reliability. The MLRF shall provide a Mean Ranges Between Failures (MRBF) of not less than 40,000 ranges. This shall be the lower test ($\theta_1$) MRBF and shall be demonstrated with 80 percent confidence. Replacement of batteries is not included in this requirement.



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3.5 Physical characteristics.

3.5.1 Weight.

3.5.1.1 MLRF weight. The weight of the MLRF excluding battery, shall not exceed 3.7 pounds.

3.5.1.2 MELIOS system weight (fielded). The weight of the MELIOS system, including the MLRF (without a battery), tripod, lens cleaning kit, and Laser Rangefinder case, excluding the shipping and storage container, 2 batteries and operator's manual shall not exceed 6.5 pounds.

3.5.1.3 MELIOS system weight (shipping). The weight of the MELIOS system (less batteries), as described in 1.1 shall not exceed 30.0 pounds.

3.5.2 Finish. Chemical and biological agent resistant materials and finishes per MIL-C-46168 shall be used to facilitate decontamination and use of the MLRF in a toxic environment. Finishes shall also be anti-reflective.

3.5.3 Marking. The assembly and all spared components, subassemblies and parts shall be marked in accordance with MIL-M-13231 as specified on the drawings.

3.5.4 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9. Soldering shall be in accordance with MIL-STD-2000. No burrs, sharp edges, roughness on welding and braze flash shall be allowed. Parts and materials shall not be damaged or missing.

3.6 Environmental stress screening (ESS). Each MLRF shall accumulate a minimum of 24 hours exposure composed of repetitions of the temperature and power profiles in Figure 2 and the vibration profile per Figure 2 and Figure 5 and with the last two cycles failure free.

3.7 Performance.

3.7.1 Beam characteristics.

3.7.1.1 Operating wavelength. All laser emissions shall be at 1.54 $\pm$ .01 microns.

3.7.1.2 Pulse width. Each MLRF shall exhibit pulse widths which fall between 16 and 30 nanoseconds inclusive and each pulse width shall equal the mean value $\bar{x} \pm 3$ nanoseconds for the MLRF tested.

3.7.1.3 Single pulse. The MLRF shall produce only one laser pulse resulting from each range operation. The MLRF shall emit this pulse only when the "FIRE" switch is depressed in accordance with the correct ranging sequence (see 3.12.5).



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3.7.1.4 Beam divergence. Not less than 80 percent of the output energy shall be contained within a 1.0 milliradian diameter beam divergence as measured in the far field.

3.7.1.5 Output energy. The output energy of each MLRF unit shall be between 10 and 33 millijoules. The pre ESS mean measurement shall be between 15 and 19 millijoules. The maximum dispersion about the mean value shall not exceed ± 25 percent of the mean value. The rate of exposure at the output aperture shall not exceed 1 $J/CM^2$.

3.7.2 Ranging.

3.7.2.1 Range capability. The MLRF shall determine and display in the eyepiece the distance to a target from a minimum of 50 meters to a maximum of 9995 meters, in 5 meter increments. Attempted rangings at targets which are at a distance of 35 meters or less and targets which are at a distance of 10,005 meters or greater shall result in a display of four zeros (0000) in the eyepiece. The MLRF shall exhibit no more than one failure out of any 100 consecutive rangings thereby determining the range to targets under the conditions that the targets shall be normal to the laser beam, the target size shall not be greater than 10 meters square, the target shall be at a range not less than 9990 meters, the target shall have diffused Lambertian reflectance of not greater than 10 percent at the laser operating wavelength, and the visual range shall be 20,000 meters.

3.7.2.2 Ranging logic. Ranging logic shall be first/last pulse, switch selectable.

3.7.2.3 Target discrimination. The MLRF shall discriminate between targets along the same line-of-sight where range separation is 50 meters or greater and shall provide positive, visual indication for multiple targets.

3.7.2.4 Range error. The range error shall not exceed ± 5 meters.

3.7.2.5 False alarm rate. The false alarm rate shall not exceed one false range reading in one hundred (100) range operations.

3.7.3 Duty cycles.

3.7.3.1 Sustained duty cycle. The MLRF shall be capable of continuous operation at a minimum rate of 6 rangings per minute.

3.7.3.2 Burst rate duty cycle. The MLRF shall be capable of continuously performing at a burst rate duty cycle consisting of 10 rangings per minute for 2 minutes followed by a 2 minute cool down period.



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3.7.4 System timing.

3.7.4.1 Charge time. The MLRF shall provide a positive visual ready-to-fire indication within 3 seconds of activation of the charge switch. The ready-to-fire status of the MLRF, to include the visual indicator in the eyepiece, shall be maintained for 8 ± 1 seconds, after which the unit shall shut down automatically.

3.7.4.2 Display time. The range shall be displayed within 1 second after activation of the fire switch. When the fire switch is released the range shall be automatically displayed for a period of 5 ± 1 seconds, after which the unit shall shut down automatically. Any time the charge switch is activated, the MLRF shall immediately cease to display the range and return to a "ready" status within 3 seconds. The range shall remain displayed when the fire switch is activated and held in the fire position.

3.7.5 System response. The MLRF shall be capable of performing a single ranging operation within 5 seconds after turning the system on.

3.7.6 System Recycling. System recycling shall be automatic between ranging sequences. The battery drain between ranging sequences shall be less than 1 milliampere.

3.7.7 Battery operation.

3.7.7.1 Battery compatibility. The MLRF shall be battery powered and shall use as its primary power source the BA-6516/U Lithium Thionyl Chloride battery (MIL-B-49461/1). The MLRF shall be capable of utilizing, as an alternate source, the BB-516/U Nickel Cadmium battery (MIL-B-49436/6). Both batteries shall be field replaceable by the operator without the use of tools.

3.7.7.2 Energy consumed per ranging operation. The MLRF shall consume less than 1200 milliampere seconds per ranging operation at an ambient temperature of 23 ± 5°C (72 ± 9°F). The ranging operation shall include charge time, fire time and display time.

3.7.7.3 Low power indicator. A positive visual indication through the eyepiece showing a low battery condition shall be provided to the operator. When the voltage is monitored at the battery terminals the indicator shall illuminate when the voltage drops to between 20.5 and 19.5 Vdc, and remain illuminated until the shutdown voltage is reached. Alternately, the low power indication shall illuminate when the voltage drops between 21.5 and 20.5 Vdc when power is applied to the test connector, due to the voltage drop of the internal protection diode.

3.7.7.4 Low power activation. The MLRF shall neither be damaged nor suffer subsequent degradation of performance when attempting to operate below the power conditions specified in 3.7.7.3. Inability to obtain a ready signal under these conditions shall not constitute a failure.



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3.7.8 Sighting optics.

3.7.8.1 Magnification. Sighting optics magnification shall be not less than 7 power.

3.7.8.2 Field-of-view. The horizontal field-of-view of the sighting optics shall be 7 degrees minimum. The vertical field-of-view shall be 5 degrees minimum.

3.7.8.3 Visible light transmission. Light transmission of the sighting optics in the visible light range shall be a minimum of 25 percent. This requirement shall be met, in its entirety, with respect to the requirement of the Classified Appendix A.

3.7.8.4 Resolution. The sighting optics of the MLRF shall be capable of resolving an object that subtends an angle of not greater than 20 arc seconds.

3.7.8.5 Reticle. The reticle shall be in accordance with A3187116. The one (1) milliradian aiming circle shall be boresighted to the transmitter and receiver.

3.7.8.6 Reticle illumination. The MLRF reticle shall have the ability to be illuminated for night and low light level operation. The reticle when not illuminated shall not cause a drain on the battery. The reticle illumination shall be actuated by a push/pull switch with intensity varied by rotation of the switch.

3.7.8.7 Optical improvements. The MLRF shall meet the requirements of 30.2.1, 30.2.2 and 30.2.3 included in the Classified Appendix A.

3.7.8.8 Display/visual indicators. All visual indicators shall be positioned in a manner such that, they can be viewed easily through the eyepiece while not reducing the verticle field-of-view to less than requirements of 3.7.8.2. All visual indicators shall be evenly illuminated and of a color and intensity such that when activated they are clearly visible in bright sunlight.

3.7.9 Eyepiece.

3.7.9.1 Diopter. The eyepiece shall provide a diopter adjustment over the range of +4 to -4.

3.7.9.2 Exit pupil. The eyepiece shall have an exit pupil diameter of not less than 5 millimeters.

3.7.9.3 Eye relief. The eyepiece shall have an eye relief of not less than 25 millimeters.



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3.7.9.4 Eye shield. The eyepiece shall be equipped with a flexible eye shield which shall reduce extraneous light. The MLRF with eye shield shall be fully functional with all military issue eyeglasses or equivalent.

3.7.10 Flashlamp stray emission. All flashlamp emissions shall be contained within the equipment housing.

3.7.11 Mounting interface alignment. The keyed mounting interface surface shall optically align with the MLRF sighting optics to within 0.25 milliradians and shall not shift more than 0.1 milliradians under any conditions.

3.7.12 Boresight.

3.7.12.1 Boresight error. The boresight error between the Laser Beam and the optical axis of the receiver/sighting telescopes shall not exceed 0.5 milliradians.

3.7.12.2 Boresight shift. Boresight shift between the Laser Beam and the optical axis of the receiver/sighting telescope shall not exceed 0.25 milliradians from the pretest boresight inspection of the unit under test while remaining within the requirements of 3.7.12.1.

3.7.13 Ancillary equipment.

3.7.13.1 Tripod. The tripod shall be in accordance with A3187238. The mounted MLRF shall remain fixed to the tripod during ranging.

3.7.13.2 Laser rangefinder case. The Laser Rangefinder case shall be in accordance with A3187229 and shall contain the MLRF, tripod, lens cleaning kit, operator's manual, and space for two BA-6516/U batteries. All items shall fit within the closed Laser rangefinder case.

3.7.13.3 Shipping and storage container. The shipping and storage container shall be in accordance with A3187239 and shall contain the Laser Rangefinder case packed with the MLRF, tripod, lens cleaning kit, operator's manual, and two BA-6516/U batteries. The shipping case shall be waterproof and protect its contents from shock and vibration while being transported as loose cargo.

3.7.13.4 Lens cleaning kit. The cleaning kit consists of one package of lens cleaning tissue (NSN 6650-00-597-6745) and a 1/2 ounce plastic dropper bottle of lens cleaning compound (NSN 6850-01-796-9656). The MLRF shall not be damaged or degraded by use of this lens cleaning kit.

3.8 Environmental requirements.

3.8.1 Temperature.



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3.8.1.1 High temperature. The MLRF shall exhibit neither damage, nor degradation of performance when operated in temperatures to +49°C (+120°F). The MLRF shall exhibit neither damage, nor degradation of performance after being stored in a non-operating mode in temperatures to +71°C (+160°F).

3.8.1.2 Low temperature. The MLRF shall exhibit neither damage, nor degradation of performance when operated in temperatures as low as -32°C (-25°F). The MRLF shall exhibit neither damage, nor degradation of performance after being stored in a non-operating mode in temperatures as low as -57°C (-70°F).

3.8.1.3 Temperature shock. The MLRF shall exhibit neither damage, nor degradation of performance after being subjected, in the non-operating mode, to sudden temperature changes between -57°C (-70°F) and +71°C (+160°F) inclusive.

3.8.1.4 Solar radiation. The MLRF shall exhibit neither damage, nor degradation of performance when subjected to temperature fluctuations from +32°C (+89°F) to +49°C (+120°F) with exposure to solar radiation.

3.8.2 Immersion. The MLRF in a non-operating mode, shall not leak when immersed under not less than 1 meter (3.3 feet) of water for not less than 120 minutes, and upon retrieval shall exhibit neither damage, nor degradation of performance during subsequent operations.

3.8.3 Humidity. The MLRF shall not experience any leakage, damage, nor degradation of performance resulting from being subjected to a relative humidity and temperature excursion as identified by Cycle 3 of Table 507.2-I and Figure 507.2-1 of MIL-STD-810.

3.8.4 Altitude.

3.8.4.1 Storage and operation. The MLRF shall exhibit neither damage nor degradation of performance due to storage or operation within altitudes up to 15,000 feet in an unprotected operating mode.

3.8.4.2 Rapid decompression. The MLRF, in the Laser rangefinder case, shall not present a safety or health hazard to personnel, or aircraft either during, or after exposure to an equivalent altitude change from 15,000 to 40,000 feet, which occurs within 15 seconds. Additionally, the MLRF shall present neither a safety nor health hazard when subsequent operation is attempted.

3.8.5 Fungus. The MLRF, Laser Rangefinder case, tripod and shipping and storage container shall not support fungus growth, nor experience damage, nor degradation of performance by the presence of fungus spores or adjacent fungus growth.



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3.8.6 Vibration.

3.8.6.1 Field vibration. The MLRF shall exhibit neither damage nor degradation of performance by exposure to a thirty (30) minute vibration, 10 minutes in each orthogonal axis.

3.8.6.2 Transportation vibration. The MLRF, in its shipping and storage container, containing all the ancillary items, shall exhibit neither damage, nor degradation of performance after being subjected to vibrations typically induced when transported as loose cargo under hostile field conditions.

3.8.7 Shock.

3.8.7.1 Field shock. The MLRF shall exhibit neither damage, nor degradation of performance resulting from physical shocks typically encountered during handling, and field use.

3.8.7.2 Transportation shock. The MLRF, in its shipping and storage container, containing all the ancillary items, shall exhibit neither damage, nor degradation of performance as a result of physical shocks incurred during typical field transportation, including loading and unloading from tracked and wheeled vehicles, and manportable transportation.

3.8.7.3 Bench handling. The MLRF shall not be damaged nor degraded in performance after experiencing mechanical shocks commonly induced during repair, maintenance, and system checkout.

3.8.8 Sand and dust. The MLRF shall exhibit neither damage nor degradation of performance by exposure, with lens covers installed, to blowing sand and dust.

3.8.9 Salt fog. The MLRF and the shipping and storage container shall exhibit neither damage nor degradation of performance due to exposure to a salt atmosphere.

3.8.10 Explosive atmosphere. The MLRF shall not initiate combustion when operated in an atmosphere that is heavily laden with fumes of aircraft or ground vehicle fuels.

3.9 Electromagnetic interference (EMI). The MLRF shall comply with the emission and susceptibility requirements of MIL-STD-461, Class A3, Table 4-1 for laser devices RE02 and RS03 only.

3.10 Nuclear survivability. The MLRF including all mission-essential mechanical configurations, optical components, electronic assemblages, electronic equipments, electronic circuits and electronic piece parts shall withstand the nuclear environments as specified in the classified Appendix A, paragraph 30.1. The MLRF shall meet the operational requirements of section



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3.7.2, of this specification after being subjected to the specified nuclear environments. The following parameters are specified criteria in Figure A-10 and A-11, and table A-3, which are included in the Classified Appendix A.

a. Time dependant total electric field (E) in volts/meter.
b. Time dependant total magnetic field (H) in amperes/meter.

c. The frequency content of the electric field (E(f)) in volts/meter/hertz.
d. Polarization.

3.11 Safety. The MLRF laser output shall be measured and the MLRF shall be classified and marked in accordance with TB-MED-524, Control of Hazards to Health from Laser Radiation dated 30 May 1975. Potentially hazardous areas within the MLRF shall be identified in accordance with MIL-STD-882. Optical products shall contain no thorium or other source material as defined by Title 10, Code of Federal Regulations Energy, Part 40, in excess of 0.05 percent by weight (500ppm) or any other added radioactive materials.

3.12 Technical interpretations. The following technical interpretations are, when referenced in sections 3, 4, or 5, mandatory for this specification.

3.12.1 Damage. Damage is the deterioration of any physical characteristic including, but not limited to, surface material, component, durability and safety or the failure of coatings, adhesives, materials, hardware and fasteners.

3.12.2 Degradation of performance. Degradation of performance is a cumulative degradation of a measured level of performance while remaining within the specified performance requirements. For reliability tests degradation of performance is the same as the limits of the specified performance requirements. For all other tests or series of tests, degradation of performance is the failure of any MLRF to perform within the specified requirements as well as the following criteria.

a. The post-ESS mean pulse width shall remain within +3/-5 nanoseconds of the pre-ESS mean pulse width value, and for all test results subsequent to post-ESS, the mean pulse width shall remain within ± 3 nanoseconds of the post-ESS mean value.

b. The post-ESS mean output energy shall remain within +3/-2 millijoules of the pre-ESS mean output energy value, and for all test results subsequent to post-ESS, the mean output energy shall remain within ± 3 millijoules of the post-ESS mean value.

c. The beam divergence measurement shall conform to the requirements of 3.7.1.4.

d. The Boresight shift shall conform to the requirements of 3.7.12.2



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3.12.3 Maximum lot size. The maximum allowable inspection lot after IPT and the first production lot shall be no greater than 300 units.

3.12.4 Successful range operation. A successful range operation is where the MLRF is operated through a ranging sequence (see 3.12.5) and displays a range reading that is within ± 5 meters of the actual target distance.

3.12.5 Ranging sequence. A ranging sequence is defined as follows:

a. Select First or Last RTN position.
b. Press the "Charge" button.
c. Wait for the "Ready" light to illuminate.
d. Press the "Fire" button.
e. Read the displayed range.

3.12.6 Qualified product. A product that has successfully met all the requirements of an IPT or First Article witnessed by the Project Manager, Night Vision and Electro-Optics or its designated representative. Should the supplier make changes in the process, component, material, personnel, or equipment used to produce a qualified product, or have not delivered the qualified product to the Federal Government within 12 months of contract award, an IPT shall be performed to the extent necessary to verify continued compliance to specification requirements for all characteristics of the product affected by the changes.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor shall be responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.



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4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection. (see 4.4).
b. Quality conformance inspection. (see 4.5).

4.3 Inspection conditions.

4.3.1 Inspection power source. Unless specifically specified all inspections/tests specified herein, shall use the BA-6516/U or BB-516/U battery or an external 24Vdc power supply as the MLRF power source.

4.3.2 Sequence of inspections. Unless otherwise specified herein inspections and tests shall be performed in any order.

4.3.3 Test conditions. Unless otherwise specified test conditions shall be in accordance with the standard ambient test conditions of MIL-STD-810.

4.4 First article inspection. When specified by the contract or purchase order, Initial Production Testing (IPT) shall be performed using a minimum of 15 systems (see 6.2.b).

4.4.1 Inspection. Each MELIOS unit shall be inspected in accordance with Table I. The presence of one or more defects shall be cause for rejection of that MELIOS unit and may be cause for the rejection of the IPT.

TABLE I. Inspections.

| Requirement | Requirement paragraph | Inspection criteria |
|---|---|---|
| Finishes not as specified | 3.5.2 | 4.11.2 |
| Marking not as specified | 3.5.3 | 4.11.3 |
| Workmanship not as specified | 3.5.4 | 4.11.4 |
| Materials and components not as specified | 3.3 | 4.11.5 |

4.4.2 IPT tests. Upon successful completion of the inspections specified in 4.4.1, each MLRF shall be subjected to the ESS requirements of 4.6, and upon successful completion, tested in accordance with Table II. For high temperature-operation, temperature shock, field vibration and field shock tests the MLRF shall contain and, if applicable, operate from a battery. Before and after each environmental test identified in Table II each test item shall be inspected in accordance with Table III Group A inspection. After completion of Subgroup 1 each MLRF shall be identified with its applicable subgroups (see 4.4.3). Failure of any test may constitute failure of IPT.



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4.4.3 Testing sequences. Upon successful completion of ESS all First Article units shall be subjected to the inspections of Table II, Subgroup 1. When Subgroup 1 inspections are completed Subgroups 2, 3, and 4 shall be performed. Six units minimum shall be assigned to Subgroup 2, three units minimum shall be assigned to Subgroup 3, and six units shall be assigned to Subgroup 4 (reliability). For the inspections of Subgroup 2 only three units are required to be subjected to the individual tests. However, the same three units shall be subjected to, Temperature Shock, Vibration, Shock, and Immersion in this order. For the inspections of Subgroup 3 marked X(1) only one unit is required to be subjected to any one test. Except for these specific sequences, all other tests may be performed in any order.

TABLE II. First article inspection.

| Requirement | Requirement Paragraph | Test Paragraph | Subgroup 1 (15) (min) | 2 (6) (min) | 3 (3) (min) | 4 (6) |
|---|---|---|---|---|---|---|
| Weight | 3.5.1 | 4.11.1 | | | X | |
| Operating wavelength | 3.7.1.1 | 4.7.1.1 | | | X | |
| Pulse width | 3.7.1.2 | 4.7.1.2 | X | | | |
| Single pulse | 3.7.1.3 | 4.7.1.3 | X | | | |
| Beam divergence | 3.7.1.4 | 4.7.1.4 | X | | | |
| Output energy | 3.7.1.5 | 4.7.1.5 | X | | | |
| Range capability | 3.7.2.1 | 4.7.2.1 | X | | | |
| Ranging logic | 3.7.2.2 | 4.7.2.2 | X | | | |
| Target discrimination | 3.7.2.3 | 4.7.2.3 | X | | | |
| Range error | 3.7.2.4 | 4.7.2.4 | X | | | |
| False alarm rate | 3.7.2.5 | 4.7.2.5 | X | | | |
| Sustained duty cycle | 3.7.3.1 | 4.7.3.1 | X | | | |
| Burst rate duty cycle | 3.7.3.2 | 4.7.3.2 | X | | | |
| Charge time | 3.7.4.1 | 4.7.4.1 | | X | | |
| Display time | 3.7.4.2 | 4.7.4.2 | | X | | |
| System response | 3.7.5 | 4.7.5 | | X | | |
| System recycling | 3.7.6 | 4.7.6 | | X | | |
| Battery compatibility | 3.7.7.1 | 4.7.7.1 | | X | | |
| Energy consumed per ranging operation | 3.7.7.2 | 4.7.7.2 | | X | | |
| Low power indicator | 3.7.7.3 | 4.7.7.3 | | X | | |
| Low power activation | 3.7.7.4 | 4.7.7.4 | | X | | |
| Magnification | 3.7.8.1 | 4.7.8.1 | | | X | |
| Field of view | 3.7.8.2 | 4.7.8.2 | | | X | |
| Visible light transmission | 3.7.8.3 | 4.7.8.3 | | | X | |
| Resolution | 3.7.8.4 | 4.7.8.4 | | | X | |
| Reticle | 3.7.8.5 | 4.7.8.5 | | | X | |
| Reticle illumination | 3.7.8.6 | 4.7.8.6 | | | X | |



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TABLE II. First article inspection. - Continued

| Requirement | Requirement Paragraph | Test Paragraph | Subgroup 1 (15) (min) | 2 (6) (min) | 3 (3) (min) | 4 (6) |
|---|---|---|---|---|---|---|
| Optical improvement | 3.7.8.7 | 4.7.8.7 | | | X | |
| Diopter | 3.7.9.1 | 4.7.9.1 | | | X | |
| Display/visual indicators | 3.7.8.8 | 4.7.8.8 | | | X | |
| Tripod | 3.7.13.1 | 4.7.13.1 | | | X | |
| Shipping and storage container | 3.7.13.3 | 4.7.13.3 | | | X(1) | |
| Laser rangefinder case | 3.7.13.2 | 4.7.13.2 | | | X(1) | |
| Lens cleaning kit | 3.7.13.4 | 4.7.13.4 | | | X | |
| Exit pupil | 3.7.9.2 | 4.7.9.2 | | | X | |
| Eye relief | 3.7.9.3 | 4.7.9.3 | | | X | |
| Eye shield | 3.7.9.4 | 4.7.9.4 | | | X | |
| Flashlamp stray emission | 3.7.10 | 4.7.10 | | X | | |
| Mounting interface alignment | 3.7.11 | 4.7.11 | | X | | |
| Boresight error | 3.7.12.1 | 4.7.12.1 | X | | | |
| Boresight shift | 3.7.12.2 | 4.7.12.2 | X | | | |
| Reliability | 3.4 | 4.5.4.4 | | | | X |
| High temperature-storage | 3.8.1.1 | 4.8.1.1.1 | | X | | |
| High temperature-operation | 3.8.1.1 | 4.8.1.1.2 | | X | | |
| Low temperature | 3.8.1.2 | 4.8.1.2 | | X | | |
| Temperature shock | 3.8.1.3 | 4.8.1.3 | | X | | |
| Solar radiation | 3.8.1.4 | 4.8.1.4 | | X | | |
| Immersion | 3.8.2 | 4.8.2 | | X | | |
| Humidity | 3.8.3 | 4.8.3 | | X | | |
| Storage and operation | 3.8.4.1 | 4.8.4.1 | | | X | |
| Rapid decompression | 3.8.4.2 | 4.8.4.2 | | | X(1) | |
| Fungus | 3.8.5 | 4.8.5 | | | X(1) | |
| Vibration | 3.8.6 | 4.8.6 | | X | | |
| Shock | 3.8.7 | 4.8.7 | | X | | |
| Sand and dust | 3.8.8 | 4.8.8 | | | X(1) | |
| Salt fog | 3.8.9 | 4.8.9 | | | X(1) | |
| Explosive atmosphere | 3.8.10 | 4.8.10 | | | X(1) | |
| Electromagnetic interference | 3.9 | 4.9 | | | X(1) | |
| Nuclear survivability | 3.10 | 4.10 | | | X(1) | |
| Safety | 3.11 | 4.11.6 | | | X(1) | |



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4.5 Quality conformance inspection. Inspection shall be as specified in Table I and shall be conducted on all MELIOS systems.

4.5.1 Group A inspection. Group A inspections listed in Table III shall be conducted on all MLRF units which have passed the inspections of Table I and ESS specified in 4.6 inclusive. Failure of any test shall constitute cause for rejection of that unit. Units successfully meeting the requirements of Group A inspection shall be formed into lots (see 3.12.3) for the purposes of B, C and D inspection.

TABLE III. Group A inspection.

| Requirement | Requirement paragraph | Test paragraph |
|---|---|---|
| Pulse width | 3.7.1.2 | 4.7.1.2 |
| Single pulse | 3.7.1.3 | 4.7.1.3 |
| Beam divergence | 3.7.1.4 | 4.7.1.4 |
| Output energy | 3.7.1.5 | 4.7.1.5 |
| Range capability | 3.7.2.1 | 4.7.2.1 |
| Ranging logic | 3.7.2.2 | 4.7.2.2 |
| Target discrimination | 3.7.2.3 | 4.7.2.3 |
| Range error | 3.7.2.4 | 4.7.2.4 |
| False alarm rate | 3.7.2.5 | 4.7.2.5 |
| Sustained duty cycle | 3.7.3.1 | 4.7.3.1 |
| Burst rate duty cycle | 3.7.3.2 | 4.7.3.2 |
| Boresight error | 3.7.12.1 | 4.7.12.1 |
| Boresight shift | 3.7.12.2 | 4.7.12.2 |

4.5.2 Group B inspection. Group B inspection shall be conducted on samples selected from lots which have passed the tests of 4.5.1. Sampling plans shall be as specified in the contract or purchase order (see 6.2.f). Group B tests listed in Table IV may be performed in any order.



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TABLE IV. Group B inspection.

| Requirement | Requirement paragraph | Test paragraph |
|---|---|---|
| Subgroup 1 | | |
| Flash lamp stray emission | 3.7.10 | 4.7.10 |
| Charge time | 3.7.4.1 | 4.7.4.1 |
| Display time | 3.7.4.2 | 4.7.4.2 |
| System response | 3.7.5 | 4.7.5 |
| System recycling | 3.7.6 | 4.7.6 |
| Battery compatibility | 3.7.7.1 | 4.7.7.1 |
| Subgroup 2 | | |
| Low power indicator | 3.7.7.3 | 4.7.7.3 |
| Low power activation | 3.7.7.4 | 4.7.7.4 |
| Mounting interface alignment | 3.7.11 | 4.7.11 |

4.5.3 Group C inspection. Group C inspection shall be conducted on samples selected from lots which have passed Group A inspection. The sample units shall be tested in accordance with the inspections listed in Table V. Tests may be performed in any order except shock and vibration shall be performed before immersion.

4.5.3.1 Sampling for Group C inspections. Sampling for Group C inspection shall be as specified in the contract or purchase order (see 6.2.f).

4.5.3.2 Group C failures. Actions regarding Group C failures shall be as specified by the contract or purchase order (see 6.2.d).

4.5.3.3 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract or purchase order provided all damage is repaired and the sample units are resubjected to, and pass, Group A inspection.



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TABLE V. Group C inspection.

| Requirement | Requirement paragraph | Test paragraph |
|---|---|---|
| Subgroup 1 | | |
| Field shock | 3.8.7.1 | 4.8.7.1 |
| Field vibration | 3.8.6.1 | 4.8.6.1 |
| Immersion | 3.8.2 | 4.8.2 |
| Subgroup 2 | | |
| High temperature-operation | 3.8.1.1 | 4.8.1.1.2 |
| Low temperature proc.I and II | 3.8.1.2 | 4.8.1.2 |
| Temperature shock | 3.8.1.3 | 4.8.1.3 |
| Subgroup 3 | | |
| Humidity | 3.8.3 | 4.8.3 |
| Subgroup 4 | | |
| Storage and operation | 3.8.4.1 | 4.8.4.1 |

4.5.4 Group D inspection. Group D inspection shall be performed on samples selected from units which have passed Group A. Testing shall be conducted in accordance with Table VI.

4.5.4.1 Sampling for Group D inspection. The reliability acceptance test shall be performed on each inspection lot. Six units shall be randomly selected from each lot.

4.5.4.2 Group D failures. Actions regarding Group D failures shall be as specified in the contract or purchase order (see 6.2.d).

4.5.4.3 Re-inspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract or purchase order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection. As a minimum, the flashlamp, PFN Capacitor, and laser rod shall be replaced.

TABLE VI. Group D inspection.

| Inspection | Requirement Paragraph | Test Paragraph |
|---|---|---|
| Reliability | 3.4 | 4.5.4.4 |



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4.5.4.4 Reliability. The MLRF including a battery or simulated battery mass, shall be tested in accordance with MIL-STD-781 using Test Plan 20-6 of MIL-HDBK-781. Failure to meet the requirements of 3.4 shall constitute failure of the reliability test. Battery failures shall not be charged against the reliability requirement. The test profiles to be used for this reliability test are shown in Figures 1 and 4. During each period of on time, the units shall be operated continuously at the sustained duty cycle. The unit shall be continuously monitored for proper operation during the reliability test. While in the chamber, the occurrence of 3 misfires within any one minute period or 7 misfires within a 48 hour period shall constitute a failure. Performance shall be verified by performing Group A tests before and as a maximum after each 48 hour test period. Upon failure, if the time to failure cannot be determined, only the time to the last successful measurement of the failed parameter shall count towards fulfilling the test plan requirements. The UUT may be powered by a hard power supply but the battery or a simulated battery containing the same size and mass must be in the system during all reliability testing.

4.6 Environmental stress screening (ESS). Prior and subsequent to ESS the MLRF shall have passed the inspections of Group A inspection, Table III, and workmanship of 3.5.4. Each MLRF shall be subjected to a minimum 24 hour ESS test period of which the last 12 hours shall be failure free. The cycle in which a failure occurs shall not be counted towards the 24 hour total. The ESS test shall be comprised of repetitions of the test cycle identified by Figure 2. The unit under test shall be vibrated in accordance with Figure 5 for a minimum of 5 minutes at both hot and cold temperature in each of the three orthogonal axis. During the periods of applied power the unit under test shall be operated at the sustained duty cycle rate and monitored for successful operation to assure positive detection of any failure at the moment of occurrence. The occurrence of 3 or more misfires detected in the chamber within any one minute period or 7 misfires within a 24 hour period shall constitute a failure. Should a failure occur during the last 12 hours the MLRF shall be reworked and retested to include all required vibration until a minimum of 12 failure free hours have been achieved. Once ESS testing has commenced, no adjustment nor rework shall be made except as required to correct a failure of the unit and return it to test.

4.7 Methods of inspection.

4.7.1 Beam characteristics.

4.7.1.1 Operating wavelength. Using a narrow band pass filter and a beamsplitter both calibrated at 1.54 microns, and two fast response radiometers verify the MLRF laser output wavelength. Position the MLRF and the beamsplitter in a manner such that, the MLRF laser output strikes the 2 radiometers. Cover the MLRF sighting/receiving optics, fire the MLRF five times and record the reading on both radiometers. Establish that the two radiometers read the same, or that the percent difference is the same for each shot. Place the narrow band pass filter between one of the radiometer detectors and beamsplitter. Cover the MLRF sighting optics. Fire the MLRF



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five times and record the readings on each radiometer. Compare the readings of the radiometers in the second set of five firings. The energy of the unfiltered beam should be equal to the energy of the filtered beam, allowing for loss through the filters and the difference between the two radiometers. If the readings are not equal, identify the wavelength(s) and intensity(s) of the other beam component(s). As an alternative, a monochrometer may be used to measure the output wavelength of the laser. Scan the wavelength of the laser light emitted from the MLRF from 0.7 to 2.0 microns. The laser radiation received for wavelengths less than 1530 nanometers or greater than 1550 nanometers shall be zero. Failure to meet the requirement of 3.7.1.1 shall constitute failure of this test.

4.7.1.2 Pulse width. Position a fast response photodiode sensitive to 1.54 micron radiation in front of the MLRF in order to receive the transmitted pulse from the MLRF. Connect the output of the photodiode to a fast response storage oscilloscope equipped with a scope camera or an equivalent recording device. Retain the photograph or record for use in 4.7.1.3. Cover the MLRF sighting optics and fire the MLRF five (5) times. Determine the pulse width at half amplitude from the stored image. Record and average the five (5) pulse widths. Failure of either the mean value or any individual value to meet the requirements of 3.7.1.2 shall constitute failure of this test.

4.7.1.3 Single pulse. Using the test results from 4.7.1.2 examine the results of each repetition for evidence of double or multiple pulsing. Sequence the range operation incorrectly and verify that the MLRF fires only when the fire button is depressed in accordance with the correct ranging sequence (see 3.12.5). Evidence of double or multiple pulsing during any single range operation, or evidence that the MLRF fires when the MLRF controls are operated in an improper sequence, or failure to meet the requirements of 3.7.1.3 shall constitute failure of this test.

4.7.1.4 Beam divergence. Using an optical bench, a radiometer calibrated for pulsed energy of 1.54 microns, an aperture plate with a 1.0 milliradian diameter hole, and a lens or mirror with sufficient aperture to accommodate the MLRF sighting/receiving optics, measure and record the beam divergence of the MLRF. Center the MLRF reticle on the approximate center of the lens. Align the radiometer to receive the reflected pulse from the mirror. Cover the MLRF sighting/receiving optics. Fire the MLRF five (5) times. Record and average the output energy of the laser. Place the aperture plate at the focal point between the lens/mirror and the radiometer. Position the aperture so that the reticle is centered on the opening in the aperture plate. Fire the MLRF as necessary and adjust the aperture plate for the maximum energy reading. Cover the MLRF sighting optics. Fire the MLRF five (5) more times, record and average the results of output energy. Divide the average of the second five firings by the average of the first five firings. Multiply the result by 100, this yields the percent of energy passing through the 1.0 milliradian aperture. Failure to meet the requirements of 3.7.1.4 shall constitute failure of this test.



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4.7.1.5 Output energy. Using a calibrated radiometer, measure the output energy of 10 consecutive shots. Failure of the MLRF output energy to be within the requirements of 3.7.1.5 shall constitute failure of this test.

4.7.2 Ranging.

4.7.2.1 Range capability. Using a detector sensitive to 1.54 micron radiation, a delay circuit capable of providing delays corresponding to ranges from 35 meters to 10,005 meters in 5 meter increments and a short pulse light emitting device which emits radiation at the MLRF wavelength, verify the MLRF range performance. The emitter shall radiate energy at the minimum sensitivity of the MLRF receiver. This minimum sensitivity shall be pre determined by ranging to a target as described in 3.7.2 or equivalent and observing the analog output of a MLRF receiver. The test set emitter's output energy shall simulate the return energy from the conditions specified in 3.7.2.1. The test set emitter shall be adjusted to produce three different intensities. (One intensity for near range, one intensity for mid-range and one intensity for long ranges). The test set timing circuit shall be capable of time delays sufficient to simulate ranging from 35 to 10,005 meters, multiple targets, and be selectable to at least 12 specific ranges that exercises all range display segments. Select and record the time delay/range setting. Fire the MLRF at the sustained duty cycle rate. The delay shall be varied to select ranges of 35, 55, 150, 500, 1000, 2500, 5000, 7500, 9000, 9500, 9990, and 10,005 meters. At each range selected the MLRF shall be fired not less than 10 times and the displayed range shall be the selected range within $\pm$ 5 meters and shall be recorded. Two or more incorrect range readings out of any 100 consecutive rangings or failure to meet the requirements of 3.7.2.1 shall constitute failure of this test.

4.7.2.2 Ranging logic. The test setup of 4.7.2.1 may be used. It must be capable of providing both a single optical pulse, and two optical pulses separated by 333 nanoseconds (50 meters). In the two pulse mode, the first pulse shall occur at the delay setting of the delay device and the second pulse shall occur 333 nanoseconds later. With the test device in the single pulse mode and the MLRF in the First Target mode, fire the MLRF into the test device. The MLRF shall indicate the proper range within $\pm$ 5 meters for the first pulse and the multitarget indicator shall be off. Repeat with test device in the two target mode. The MLRF should display the same range with the multitarget indicator illuminated. Switch the MLRF to the last target mode, and fire the MLRF into the test device. The MLRF should display a range 50 $\pm$ 5 meters greater than the range delay selected. The multi-target indicator shall be illuminated. Switch the test device back to the one target mode, and keep the MLRF in the last target mode. Fire the MLRF into the test device. The MLRF should display the same range as that selected on the test device, and the multitarget indicator should not be illuminated. Failure of the MLRF to give the indicated displays, or meet the requirement of 3.7.2.2 shall constitute failure of this test.



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4.7.2.3 Target discrimination. Review the results of the Ranging logic test (see 4.7.2.2). Failure of the MLRF to provide the correct display, or meet the requirement of 3.7.2.3 shall constitute failure of this test.

4.7.2.4 Range error. Review the results of the Range Capability Test (see 4.7.2.1). Two or more incorrect rangings out of any 100 consecutive rangings, or failure to meet the requirements of 3.7.2.4 shall constitute failure of this test.

4.7.2.5 False alarm rate. Using the test set up of 4.7.2.1, set the time delay between laser firing and returned emitter pulse to be equivalent to 15,000 meters. This simulates covering the receiving optics. Fire the MLRF 20 times, at the burst rate duty cycle followed by a two (2) minute cooldown period followed by another 20 shots at burst rate with another 2 minute cool down period, and 60 firings at the sustained duty cycle. Verify the laser fired each time. Record the value of the range for each time the laser fired. The range reading shall be 0000 each time the laser fired. Two or more readings other than 0000 or failure to meet the requirements of 3.7.2.5 shall constitute failure of this test.

4.7.3 Duty cycles.

4.7.3.1 Sustained duty cycle. Review the results of 4.7.2.1. Operation at the specified repetition rate of 6 ranges per minute shall demonstrate the sustained duty cycle. Failure to meet the requirements of 3.7.3.1 shall constitute failure of this test.

4.7.3.2 Burst rate duty cycle. Review the results of 4.7.2.5. Operation at the specified repetition rate of 10 ranges per minute for 2 minutes, with a 2 minute cool down period followed by another 20 shots at burst rate followed by another 2 minute cool down period shall demonstrate the burst rate duty cycle. Failure of the laser to fire during any range operation shall constitute failure of this test.

4.7.4 System timing.

4.7.4.1 Charge time. Using a time measuring device capable of resolving 0.1 second intervals, measure the time required for the system to indicate a fully charged status. Measure the length of time that the ready-to-fire indicator is maintained before automatic shutdown. Failure to meet the requirements of 3.7.4.1 shall constitute a failure of this test.

4.7.4.2 Display time. Measure the length of time from activation of the "FIRE" switch to the display of the range. Measure the time that the range continues to be displayed after the "FIRE" switch is released. Perform the ranging operation again and verify that the display remains illuminated while holding the "FIRE" switch down for a period of one minute. Perform the



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ranging operation a third time and depress the "CHARGE" button within four seconds after the range is displayed and verify that the system returns to a ready-to fire status within 3 seconds. Failure to meet any requirements of 3.7.4.2 shall constitute failure of this test.

4.7.5 System response. Using a time measuring device capable of resolving 0.1 second intervals, measure the time from the system turn-on to range display. Turn the MLRF on and begin timing. Press the charge button. When the ready light illuminates, press the FIRE button. When the range is displayed, stop the timer. Failure to meet the requirements of 3.7.5 shall constitute failure of this test.

4.7.6 System recycling. Initiate the ranging sequence of the MLRF (see 3.12.5). Note and record that the MLRF has ranged and release the buttons depressed for the ranging sequence. Note and record that the range display shuts off within 5 ± 1 seconds. Initiate the ranging sequence again. Note and record that the MLRF ranges a second time. Using the test set up of 4.7.7.2 verify that there is no battery drain between ranging operations in excess of 1 milliampere. Failure to meet the requirements of 3.7.6 shall constitute failure of this test.

4.7.7 Battery operation.

4.7.7.1 Battery compatibility. Install a BA-6516/U in the MLRF without the use of tools. Verify that the unit functions properly by performing a successful range operation (see 3.12.4). Repeat using a BB-516/U. Failure to meet the requirements of 3.7.7.1 shall constitute failure of this test.

4.7.7.2 Energy consumed per ranging operation. Energize the MLRF from a 24 ± 1 volt external supply through a precision 0.1 ohm current sampling resistor in the return lead, attached to the test connector. Connect a storage oscilloscope across the sampling resistor. Perform a ranging sequence (see 3.12.5). Record the oscilloscope display retaining the photograph or record for use. By dividing the current waveform into more than ten parts, calculate the current and time product of each part from the oscilloscope waveform in milliampere seconds. Add the parts together to derive the total milliampere seconds. Failure to meet the requirements of 3.7.7.2 shall constitute failure of this test.

4.7.7.3 Low power indicator. Energize the MLRF through the battery terminals or through the test connector from an external power source adjusted to 22Vdc applied to the battery terminal or to pin 7 of the test connector with the return connected to pin 3. Decrease the input voltage in 0.2 volt increments, cycling the MLRF through a ranging sequence at each incremental voltage level. Record the voltage level at which the low power indicator (blinking display) is activated. Failure to meet the requirements of 3.7.7.3 shall constitute failure of this test.



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4.7.7.4 Low power activation. Continue with the test set-up of 4.7.7.3, lower the voltage until the MLRF stops performing. Increase the voltage to 24Vdc. Perform a ranging operation. Failure of the MLRF to operate or meet the requirements of 3.7.7.4 shall constitute failure of this test.

4.7.8 Sighting optics.

4.7.8.1 Magnification. Measure the diameter of the entrance and exit pupil of the MLRF. The ratio of entrance to exit pupil size is the magnification. Failure to meet the requirements of 3.7.8.1 shall constitute failure of this test.

4.7.8.2 Field-of-view. Mount the MLRF on a fixture which provides an azimuth readout accurate to the nearest minute. Position the MLRF so that a convenient vertical object is at the left edge of the field of view of the sighting optics. Record the azimuth. Traverse the MLRF until the object is at the right edge of the field of view. Record the azimuth. The difference in azimuth is the horizontal field of view. Reposition the MLRF on its side. Repeat along the vertical axis of the sighting optics. The difference in the readings is the vertical field of view. Failure to meet the requirement specified in 3.7.8.2 and the vertical field of view portion of 3.7.11 shall constitute failure of the test.

4.7.8.3 Visible light transmission. Use a collimated beam of visible light with a limiting aperture. The aperture should be just slightly smaller than the aperture of the MLRF's objective lens. Place an integrating sphere so that all of the light passing through the aperture enters one of its ports. Using the photopic filter measure the intensity of the light in the sphere. Position the MLRF so that light passing the aperture passes through the MLRF and then enters the sphere. Again measure the intensity in the sphere. Calculate transmission by dividing the latter reading by the former. Failure to meet the requirement specified in 3.7.8.3 shall constitute failure of this test.

4.7.8.4 Resolution. Mount a USAF 1951 resolution target at the focal plane of a collimator. Using a dioptometer, view the target through the MLRF. Calculate the pattern which corresponds to a 20 arc seconds with the focal length of the collimator used. Verify that the sighting optics will resolve this target. Failure to meet the requirements of 3.7.8.4 shall constitute failure of this test.

4.7.8.5 Reticle. Visually inspect the MLRF reticle. Failure to meet requirement 3.7.8.5 shall constitute failure of this test.

4.7.8.6 Reticle illumination. Using the test set-up of 4.7.7.2 energize the MLRF through the test connector, charge the system, view the reticle illumination in a dark or low light environment by turning the mode switch to FIRST RTN position and activate the reticle . The reticle shall be



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visible and readable. Using the current measurement method described in 4.7.7.2, verify that the reticle illumination does not consume battery energy between ranging operations with the reticle not activated. Failure to meet the requirements as specified in 3.7.8.6 shall constitute failure of this test.

4.7.8.7 Optical improvement. See 40.1 and 40.2 of the classified Appendix A for conformance requirements. Failure to meet the requirements of 3.7.8.7 shall constitute failure of this test.

4.7.8.8 Display/visual indicators. With the MLRF in the test mode activate all the display segments. Verify that all display segments are visible and legible and appear to be equally illuminated. Conduct the test in bright sunlight. Failure to meet the requirement of 3.7.8.8 shall constitute failure of this test.

4.7.9 Eyepiece.

4.7.9.1 Diopter. Use a dioptometer or equivalent instrument. Focus the dioptometer to its reticle and set the scale to zero diopter. Place the dioptometer on the eyepiece of the Optics Module. Move the diopter ring of the Optics Module clockwise as far as it will adjust. Refocus the dioptometer on the Optics Module reticle. Read and record the scale on the dioptometer. Move the diopter ring on the Optics Module counter clockwise as far as it will adjust. Again, refocus the dioptometer on the reticle of the Optics Module. Read and record the scale on the dioptometer. Failure to meet the requirements of 3.7.9.1 shall constitute failure of the test.

4.7.9.2 Exit pupil. Measure the diameter of the exit pupil, using the reticle in a dynameter K and E Model 71-7105 or equivalent. Failure to meet the requirement of 3.7.9.2 shall constitute failure of this test.

4.7.9.3 Eye relief. Without disturbing the setting used for the previous test (Exit Pupil 4.7.9.2), determine the eye relief by reading the focus scale on the side of the dynameter. Calculate the eye relief by adding to this value the distance from the top surface of the eyepiece lens retaining ring to the top surface of the eyepiece lens. Failure to meet the requirement of 3.7.9.3 shall constitute failure of this test.

4.7.9.4 Eye shield. Perform a ranging operation with the MLRF using a chemical, biological, and radiological protective mask and perform using standard eye glasses. Failure to meet the requirement of 3.7.9.4 shall constitute failure of this test.

4.7.10 Flashlamp stray emission. Insert an opaque plug in the MLRF eyepiece to block radiation from the range display. Aim the MLRF at a diffuse black surface. Position three observers which have been dark adapted (see 3.12.7) so that they do not view the output window surface of the MLRF nor the reflecting surface at which it is fired. The area in which the test is conducted shall be a darkened room. Each of the three observers shall view



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the firing of the MLRF at a distance of 5 ± 1 meters with the naked eye and using the AN/PVS-5 Night Vision Goggle or an equivalent. Failure to meet the requirements of 3.7.10 shall constitute failure of this test.

4.7.11 Mounting Interface Alignment. Align the elements so that the target is on the center line of the collimator and the V mount is aligned parallel to this center line. In addition the V mount must be positioned such that a MLRF mounted on it will View the collimated image of the target. Measure the displacement of the target with respect to the MLRF reticle. Failure to meet the requirements of 3.7.11 shall constitute failure of this test.

4.7.12 Boresight.

4.7.12.1 Boresight error. Align the receiver/sighting optics of the MLRF with the aperture fixture and lens. With the aperture fixture at the focal point of the lens insert a piece of exposed Polaroid film marked with a cross. Align the reticle pattern of the MLRF with the reticle pattern on the film. Cover the MLRF objective lens and fire the MLRF to burn a spot on the film. Measure the distance from the center of the burn spot to the center of the reticle pattern on the film and convert the result to milliradians. Conduct this test a total of (5) five times. Failure of the results of each firing to meet the requirement of 3.7.12.1 shall constitute failure of this test.

4.7.12.2 Boresight shift. Examine the data of 4.7.12.1 taken before and after each environmental test. Determine and identify the geometric location of the average pretest and posttest boresight error. Compare the data and calculate any difference in boresight between the pretest data and the post test data. Failure to meet the requirement of 3.7.12.2 shall constitute failure of this test.

4.7.13 Ancillary equipment inspection.

4.7.13.1 Tripod. Mount the MLRF on the tripod. Inspect to verify that the tripod adjustment remains fixed through at least 5 ranging operations. The inability of the tripod to support the MLRF, loosening of any tripod adjustment during the ranging operations or failure to meet the requirement of 3.7.13.1 shall constitute failure of this test.

4.7.13.2 Laser rangefinder case. Pack the Laser Rangefinder case with all of the items required as identified in 3.7.13.2 and inspect to assure fit of all items. Inspect to assure proper and complete closure of the case. The inability of the case to accommodate the required items, or the inability to pass any required environmental test or failure to meet the requirements of 3.7.13.2 shall constitute failure to meet the requirement.



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4.7.13.3 Shipping and storage container. Remove and replace the MLRF and the ancillary equipment from the container. Failure of the shipping container to provide a secure fit, failure of the shipping container to pass any required environmental test or failure to meet the requirement of 3.7.13.3 shall constitute failure of the requirement.

4.7.13.4 Lens cleaning kit. Perform the lens cleaning procedure for the MLRF by wetting the cleaning tissue with the solvent and wiping the lenses as described in TM 11-5860-202-10. Inspect to assure that the MLRF has not been damaged or degraded by the cleaning operation and perform a range operation. Evidence of damage or degradation of performance to the MLRF as the result of the cleaning operation shall constitute failure of the MLRF to meet the requirement of 3.7.13.4.

4.8 Environmental tests. For applicable definitions of damage and degradation of performance, see 3.12.1 and 3.12.2. For the purpose of environmental testing, power source failures shall be documented as non-relevant failures. The workmanship inspection of Table I and the Group A inspections listed in Table III shall be performed before and after each individual environmental test. The tests performed after an environmental test may serve as the tests performed before the next environmental test providing all of the following are met:

a. Not more than 72 hours has elapsed since the last tests were completed.
b. The unit under test has been secured and not tested or operated during that period.
c. The previous tests revealed no failures.

4.8.1 Temperature.

4.8.1.1 High temperature.

4.8.1.1.1 High temperature-storage. The MLRF shall be subjected to the tests of MIL-STD-810, Method 501.2, Procedure I. The objective lens shall be protected during this test. During the tests the Hot Cycle with Induced Conditions as identified in Table 501.2-I shall be used. Failure to meet the requirements of 3.8.1.1 and 4.8 shall constitute failure of this test.

4.8.1.1.2 High temperature-operation. The MLRF shall be subjected to the High temperature profile of Figure 6 with the following conditions. All temperature changes shall be at 1.5°C/minute minimum. For these tests the temperatures of paragraph 3.8.1.1 shall be used respectively. The profile shall be conducted three (3) times. The operating temperature boresight shift test shall be conducted in the third cycle. At the operating hot temperature the MLRF shall be fired into burn paper or equivalent. At the conclusion of the tests the Unit Under Test shall remain at Room Ambient Temperature until stabilized prior to post testing per 4.8. Failure to burn a spot and meet the requirements of 3.8.1.1 and 4.8 shall constitute failure of this test.



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4.8.1.2 Low temperature. The MLRF shall be subjected to the tests of MIL-STD-810, Method 502.2, Procedures I, II, and III. For these tests the temperatures of 3.8.1.2 shall be used respectively. Low temperature storage period shall be a minimum of 24 hours. At the operating cold temperature the MLRF shall be fired into burn film or equivalent. The operating temperature boresight shift test shall be conducted during the operating cold temperature. Failure to burn a spot and meet the requirements of 3.8.1.2 and 4.8 shall constitute failure of this test.

4.8.1.3 Temperature shock. The MLRF shall be subjected to the test of MIL-STD-810, Method 503.2, Procedure I, in accordance with the temperatures of 3.8.1.3. During the performance of this test no diurnal cycling shall be performed. Instead, the temperature chambers shall be maintained at the constant temperatures identified in 3.8.1.3 and the MLRF shall remain in each chamber until it has stabilized. Failure to meet the requirements of 3.8.1.3 and 4.8 shall constitute failure of this test.

4.8.1.4 Solar radiation. The MLRF shall be subjected to the test of MIL-STD-810, Method 505.2, Procedure I, in accordance with the Hot-Dry Cycle identified herein (see 3.8.1.4). During the solar radiation test the MLRF shall be fired during each diurnal cycle. Failure to burn a spot into the burn film on each firing or meet the requirements of 3.8.1.4 and 4.8 shall constitute failure of this test.

4.8.2 Immersion. The MLRF shall be subjected to the test of MIL-STD-810, Method 512.2, Procedure I. Any evidence of moisture found in the unit or the battery compartment, or failure of any post-test inspection, shall constitute failure of this test.

4.8.3 Humidity. The MLRF shall be subjected to the test of MIL-STD-810, Method 507.2, Procedure I for ten (10) cycles of natural, cyclic high humidity in accordance with Table 507.2-I, Cycle 3. Failure to meet the requirements of 3.8.3 and 4.8 shall constitute failure of this test.

4.8.4 Altitude.

4.8.4.1 Storage and operation. The MLRF shall be subjected to the test of MIL-STD-810, Method 500.2, Procedure II. In Step 2, the chamber pressure shall be adjusted to an ambient air pressure equivalent to a 15,000 foot altitude, at an altitude rate of change between 1,500 ft/min and 2,000 ft/min. In Step 3, the MLRF laser shall be fired a minimum of three times, and for each firing the laser emissions shall be confirmed. Then the 15,000 foot equivalent altitude pressure shall be maintained for 1 hour without MLRF operation. At the end of the 1 hour period the MLRF laser shall be fired another three times, at a minimum, and laser emissions shall be confirmed. At the completion of these firings proceed to Step 4. A complete inspection shall be performed prior to the rapid decompression test. The absence of



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laser emissions during any laser firing performed in Step 3, or evidence of damage or degradation of performance found either during the test or during the post-test inspection, or failure to meet the requirement of 3.8.4.1 shall constitute failure of this test.

4.8.4.2 Rapid decompression. The MLRF with battery and ancillary equipment, excluding the shipping and storage container, shall be subjected to the test of MIL-STD-810, Method 500.2, Procedure III. In Step 1 the MLRF with battery installed, and all ancillary equipment shall be packed in the laser rangefinder case as it would be when transported by an individual soldier. In Step 2, the equivalent altitude shall be 15,000 feet. In Step 3, an equivalent altitude of 40,000 feet shall be used. Failure of the laser rangefinder case to fully contain all items throughout the test, or evidence of a safety or health hazard either during the test, or during post-test inspection shall constitute failure to meet the requirement of 3.8.4.2.

4.8.5 Fungus. The MLRF, Laser Rangefinder case, tripod, and shipping and storage container shall each be subjected to the test of MIL-STD-810, Method 508.3. The test duration shall be 28 days. Evidence that any test item supports the inoculant fungus growth, or any evidence of damage to test items, degradation of performance of the test items during the post-test inspections or failure to meet the requirements of 3.8.5 and 4.8 shall constitute failure of this test.

4.8.6 Vibration.

4.8.6.1 Field vibration. The MLRF with a battery or dummy mass installed shall be subjected to thirty minutes of random vibration conducted in accordance with MIL-STD-810, Method 514.3, Procedure I, using the limits of Figure 4 of this specification. The test item shall be subjected to 10 minutes of vibration in each orthogonal axes, for a total time of thirty minutes. Failure to meet the requirements of 3.8.6.1 and 4.8 shall constitute failure of this test.

4.8.6.2 Transportation vibration. The MELIOS shall be subjected to the test of MIL-STD-810, Method 514.3, Procedure II. Failure to meet the requirements of 3.8.6.2 and 4.8 shall constitute failure of this test.

4.8.7 Shock.

4.8.7.1 Field shock. The MLRF with a battery or dummy mass installed shall be subjected to 3 shock pulses in each direction of each of the three (3) orthogonal axes, for a total of 18 shock pulses, using an a suitable shock machine. The MLRF shall be securely mounted by the V block mounting point to a transponder and the shock pulses shall be measured from a point on the MLRF as close to the mounting point as possible. The applied shock pulses shall be of a terminal peak sawtooth shock pulse configuration as detailed in Figure 3 of this specification. Failure to meet the requirements of 3.8.7.1 and 4.8 shall constitute failure of this test.



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4.8.7.2 Transportation shock. The MELIOS shall be subjected to the tests of MIL-STD-810, Method 516.3, Procedure IV. The test height shall be not less than 48 inches from the point of release to the point of impact. Evidence of damage or degradation of performance of the MLRF detected during the test or upon post test inspection shall constitute failure of this test. Damage to the MELIOS shipping and storage container shall be recorded and reported in the test report, but shall not be considered as a failure unless such damage renders the shipping case ineffective in protecting the MLRF, and/or the shipping and storage container is damaged in such a manner as to preclude removal of the MLRF by one person without the use of tools. Failure to meet the requirements of 3.8.7.2 and 4.8 shall constitute failure of this test.

4.8.7.3 Bench handling. The MLRF shall be subjected to the test of MIL-STD-810, Method 516.3, Procedure VI. Failure to meet the requirements of 3.8.7.3 and 4.8 constitute failure of this test.

4.8.8 Sand and dust. The MLRF shall be subjected to the tests of MIL-STD-810, Method 510.2 Procedures I and II. These tests shall be performed with all lens caps installed. For Procedure I, either red china clay or silica flour shall be used, an air velocity of 8.9 m/s (1750 ft/min) shall be used in Steps 2 and 7, and a temperature of +49°C (+120°F), shall be used in Step 5. For procedure II, the temperature shall be +49°C (120°F), the air velocity shall be 18 to 29 m/s (3540 to 5700 ft/min), and a sand concentration of 2.2 ± 0.5 g/m3 (0.0623 ± 0.015 g/ft3) shall be used. The applicable duration of exposure for Step 4 shall be 90 minutes per face of the MLRF. The MLRF shall not be operated during performance of Procedures I and II. For these tests only, degraded finish (excluding covered optical surfaces) shall be permissible, with the exception that operational and safety markings shall not be obscured to the extent of illegibility. Damage, degradation of performance, or failure to meet the requirements of 3.8.8 and 4.8, shall constitute failure of this test.

4.8.9 Salt fog. The MLRF and shipping and storage container shall each be subjected to the test of MIL-STD-810, Method 509.2, Procedure I. Failure to meet the requirements of 3.8.9 and 4.8 shall constitute failure of this test.

4.8.10 Explosive atmosphere. The MLRF shall be subjected to the test of MIL-STD-810, Method 511.2, Procedure I. In Step 3 the ambient temperature shall be +49°C (120°F), and in Step 4 the test altitude shall be 15,000 feet. The MLRF shall be configured for normal operation, and the atmosphere of the test chamber shall contain fuel weight calculated to be 3.8 percent by volume of the test atmosphere. The MLRF, when operated as specified in Step 7, shall not ignite the atmosphere in the test chamber. Provisions shall be made for monitoring operation of the MLRF while in the test chamber. Evidence of combustion (e.g. smoke, sparks, flames, or explosion) ignited by the operation of the MLRF, evidence of damage or degradation of performance, or failure to meet the requirements of 3.8.10 and 4.8 shall constitute failure of this test.



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4.9 Electromagnetic interference (EMI) tests. The MLRF shall be tested for compliance with the requirements of MIL-STD-461, Class A3, Table 4-1 for laser devices, RE02 and RS03. The MLRF shall be tested using MIL-STD-462. The MLRF shall be battery powered with the objective lens and laser window covered. During the radiated emissions tests, the MLRF shall be fired at the sustained duty cycle rate while the receiver is recording the amplitude of the emissions. During the radiated susceptibility tests, the MLRF shall be fired at the sustained duty cycle rate while observing the display. The radiated fields shall not cause the MLRF to malfunction as observed by a range other than 0000. Any range readout other than 0000 or MLRF emissions greater than the limits of MIL-STD-461 or failure to meet the requirements of 3.9 shall constitute failure of this test.

4.10 Nuclear survivability. The MLRF shall be placed within the radiation volume of a radiated-wave, ground effects EMP simulator. The MLRF shall be subjected to ten (10) EMP pulses, five (5) with the mode switch in first RTN and five with the mode switch OFF. Three pulses of each of these sets shall be done with the MLRF orientated in the three orthogonal directions. The remaining two pulses of each of these sets shall be done with the MLRF mounted on the tripod and orientated in the two horizontal directions. Before and after each pulse, the MLRF shall be tested. This test shall consist of performing a successful range operation to a target. Failure of the MLRF to obtain a range or meet the requirements of 3.10 shall constitute failure of this test.

4.11 Physical characteristics.

4.11.1 Weight.

4.11.1.1 MLRF weight. Using a calibrated balance scale or equivalent instrument, weigh the MLRF, without battery installed. Record the weight. A weight exceeding the requirement of 3.5.1.1 shall constitute a failure of this test.

4.11.1.2 MELIOS system weight (fielded). Using a calibrated balance scale or equivalent instrument weigh the MLRF (without a battery), tripod, lens cleaning kit, and Laser Rangefinder case. Record the weight. A weight exceeding that required by 3.5.1.2 shall constitute failure for this test.

4.11.1.3 MELIOS system weight (shipping). Using a calibrated balance scale or equivalent instrument weigh the MELIOS system, as defined in 1.1. A weight exceeding that required by 3.5.1.3 shall constitute failure of this test.

4.11.2 Finish. Inspect finishes and applicable certifications as necessary, for compliance with the requirements of 3.5.2. Failure to meet the requirements 3.5.2 shall constitute failure of this test.



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4.11.3 Marking. Inspect the MELIOS system for compliance with the requirements of 3.5.3 and MIL-M-13231. Failure to meet the requirements of 3.5.3 shall constitute failure of this test.

4.11.4 Workmanship. Inspect the MELIOS system for compliance with the requirements of 3.5.4. Failure to meet the requirements of 3.5.4 shall constitute failure of this test.

4.11.5 Parts, materials and processes. Assure that materials and components meet their specific requirements.

4.11.6 Safety. Inspect the MLRF and applicable certifications to verify that the optical elements, radiation markings and potentially hazardous areas within the MLRF meet the requirements of 3.11. Failure to meet these requirements of 3.11 shall constitute failure of this test.

5. Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specific by the acquisition activity in the contract or purchase order.

6. NOTES

6.1 Intended Use. The Miniature Eyesafe Laser Infrared Observation Set (MELIOS), AN/PVS-6 is intended to provide the soldier with accurate range information to any selected target.

6.2 Acquisition requirements. Procurement documents shall specify the following:

a. Title, number, and date of this specification.
b. Initial Production Test required.
c. Time frame required for submission of IPT reports.
d. Actions required relative to Group C and Group D failures (See 4.5.3.2 and 4.5.4.2).
e. The contracting officer should arrange to furnish the property listed in 4.7.7.1, 4.7.9.4 and 4.7.10.
f. Sampling plans for Group B and Group C testing. Unless otherwise specified sampling shall be in accordance with MIL-STD-105E.
g. Unless otherwise specified, MIL-STD-810D shall be used for the environmental tests of Section 4.
h. Issue of DODISS to be cited in the solicitation and if required the specific issue of individual documents required. (see 2.1.1 and 2.1.2).

6.3 Definitions. See 3.12.



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6.4 Qualification. With respect to products requiring qualification, unless otherwise specified by the contract or purchase order, if a product is not qualified at time of award, the contract shall require qualification prior to first delivery. The activity responsible for approving qualified products is the Project Manager, Night Vision and Electro-Optics, ATTN; Manager, Product Assurance and Test, Fort Belvoir, VA 22060-5677. Information pertaining to qualification of products may be obtained from the above activity.

6.5 Subject term (Keyword) listing.
MELIOS
Eyesafe Laser
Laser Rangefinder
Rangefinder
MLRF

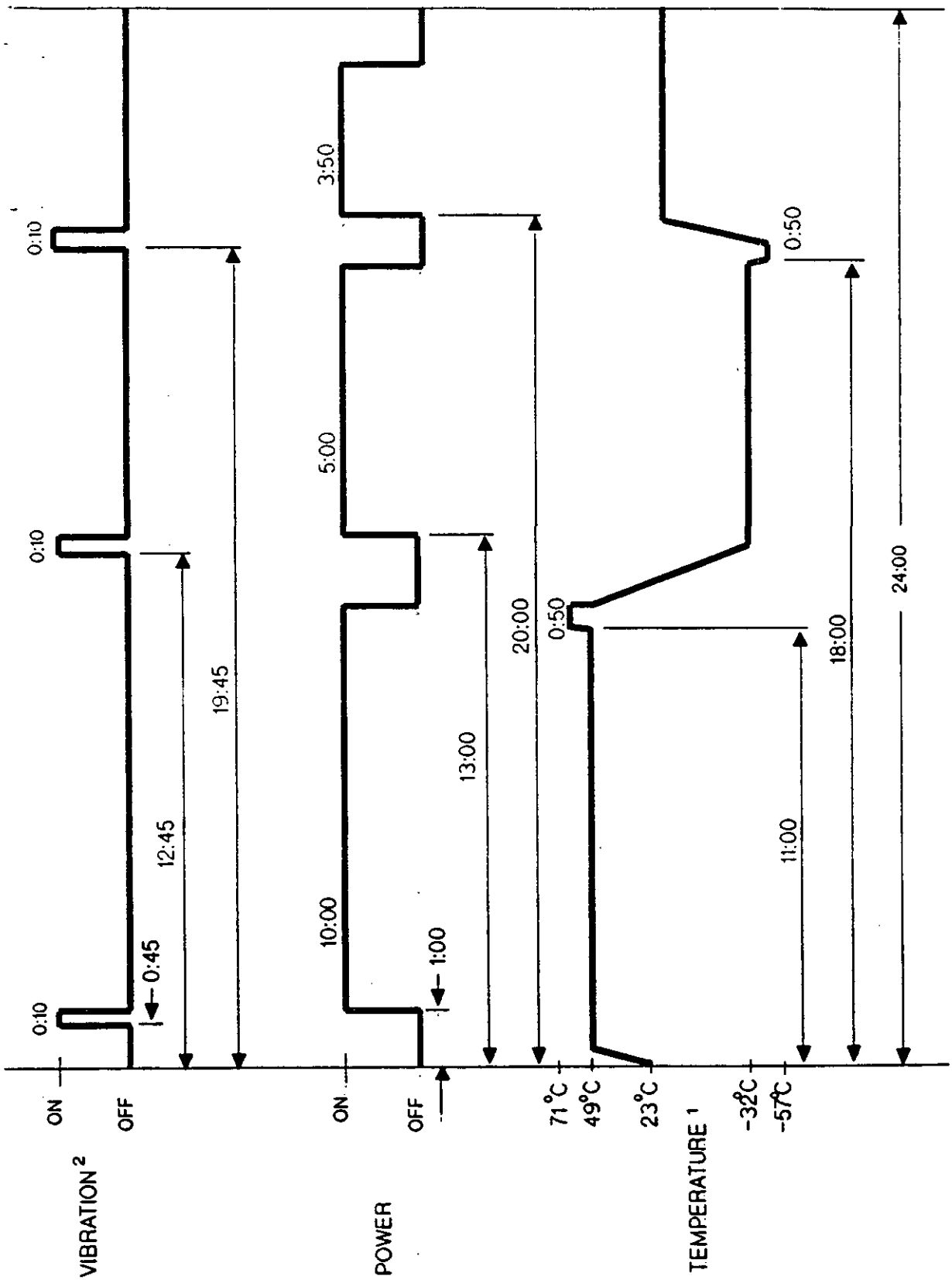
Custodian
Army-CR

Preparing Activity
Army-CR

Project 5860-A010



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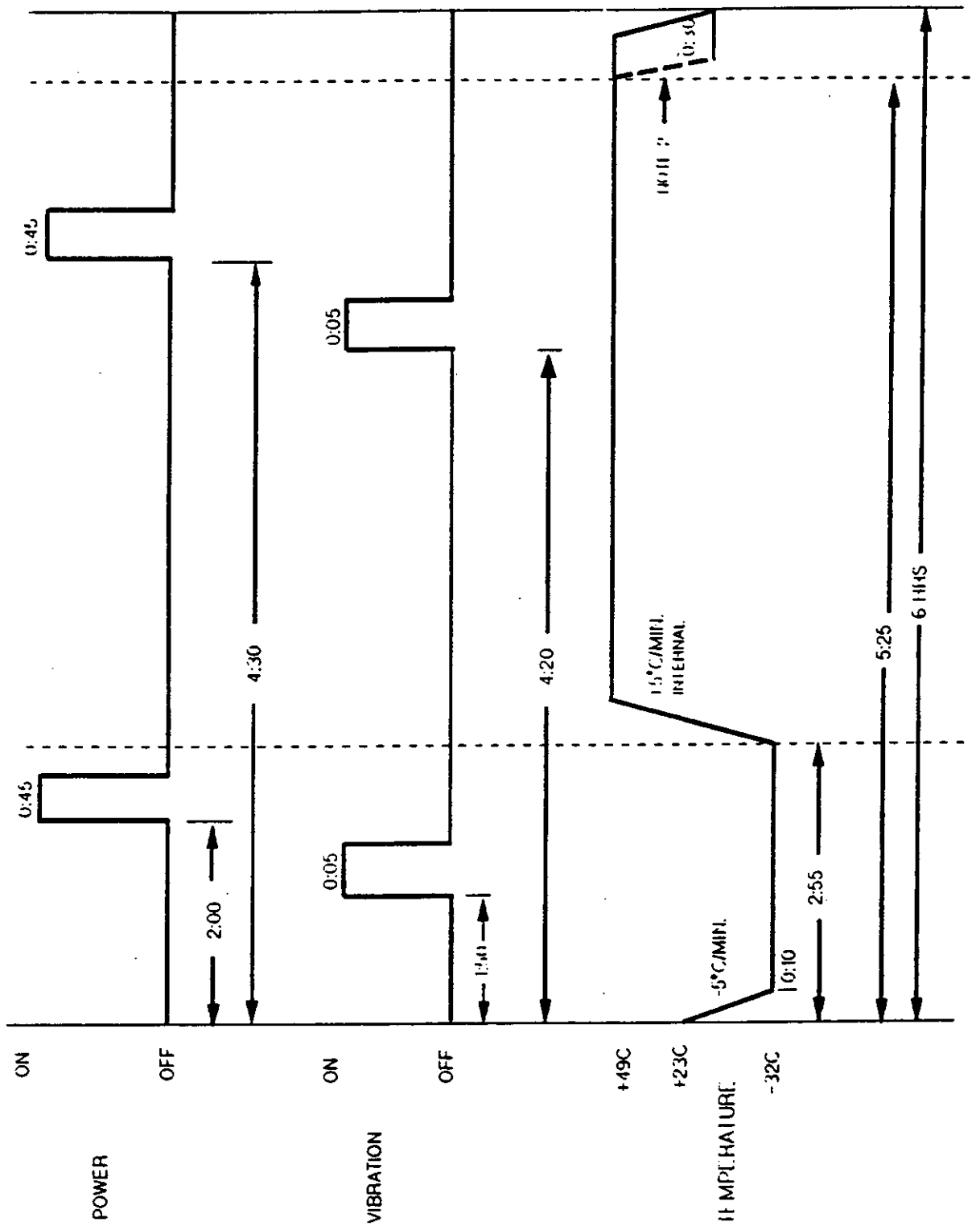

NOTE 1: All chamber temperature changes are at 5°C/minute minimum.

NOTE 2: Vibration shall be conducted equally on each of three orthogonal axes during the test period in accordance with Figure 4.

**Figure 1. Reliability Test Profile**



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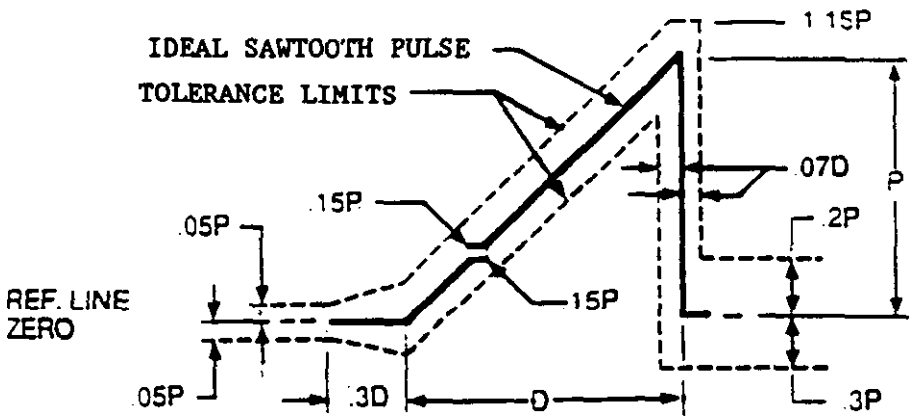

NOTE 1: Vibration shall occur in the conditions shown, once every four (4) cycles for each axis.

NOTE 2: (Dotted line) Ramp down 30 minutes early during last cycle only.

Figure 2. ESS Stress Profile



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MINIMUM PEAK VALUE (P) = 40 g's

NOMINAL DURATION (D) = 11ms

NOTE: The oscillogram shall include a time about 3D long with a pulse located approximately in the center. The peak acceleration magnitude of the sawtooth pulse is P and its duration is D. The measured acceleration pulse shall be contained between the broken line boundaries and the measured velocity change (which may be obtained by integration of the acceleration pulse) shall be within the limits of $V_1 \pm 0.1\ V_1$, where $V_1$ is the velocity change associated with the ideal pulse which equals 0.5 DF. The integration to determine velocity change shall extend from 0.4D before the pulse to 0.1D after the pulse.

Figure 3. Shock



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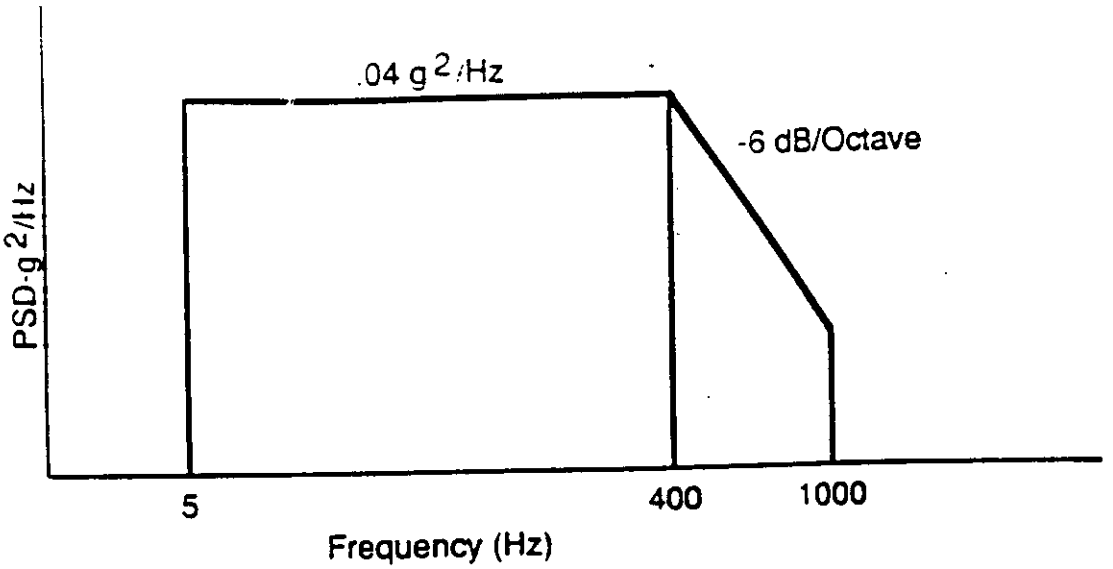



Figure 4. Reliability Profile

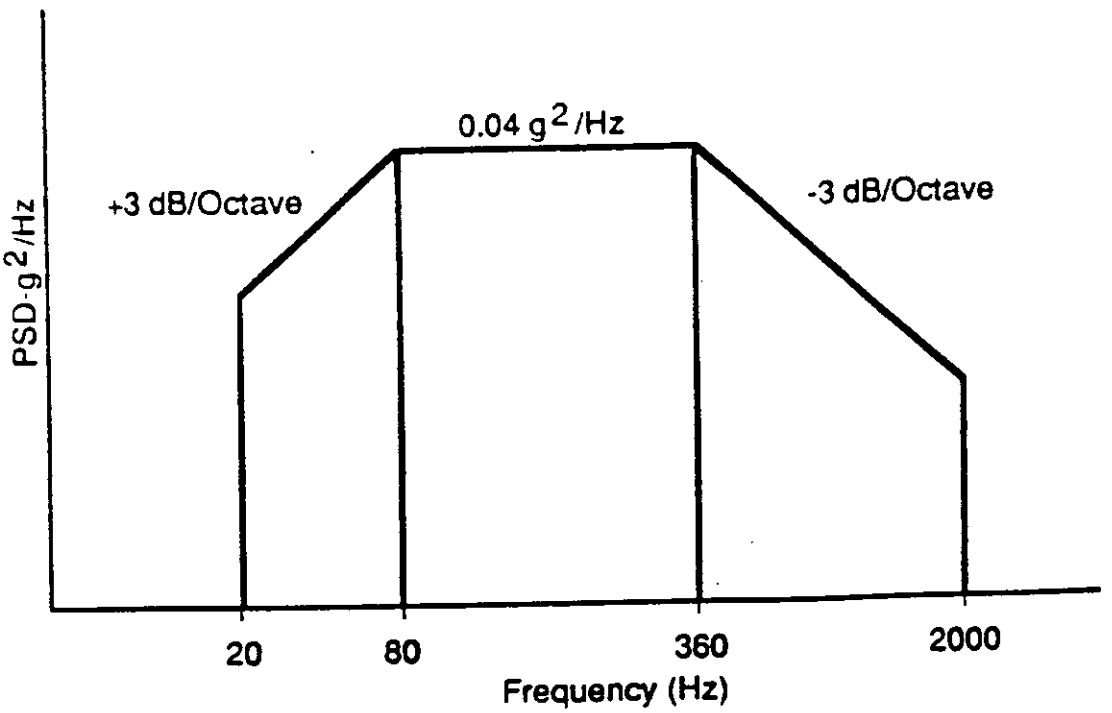



Figure 5. ESS Vibration Profile



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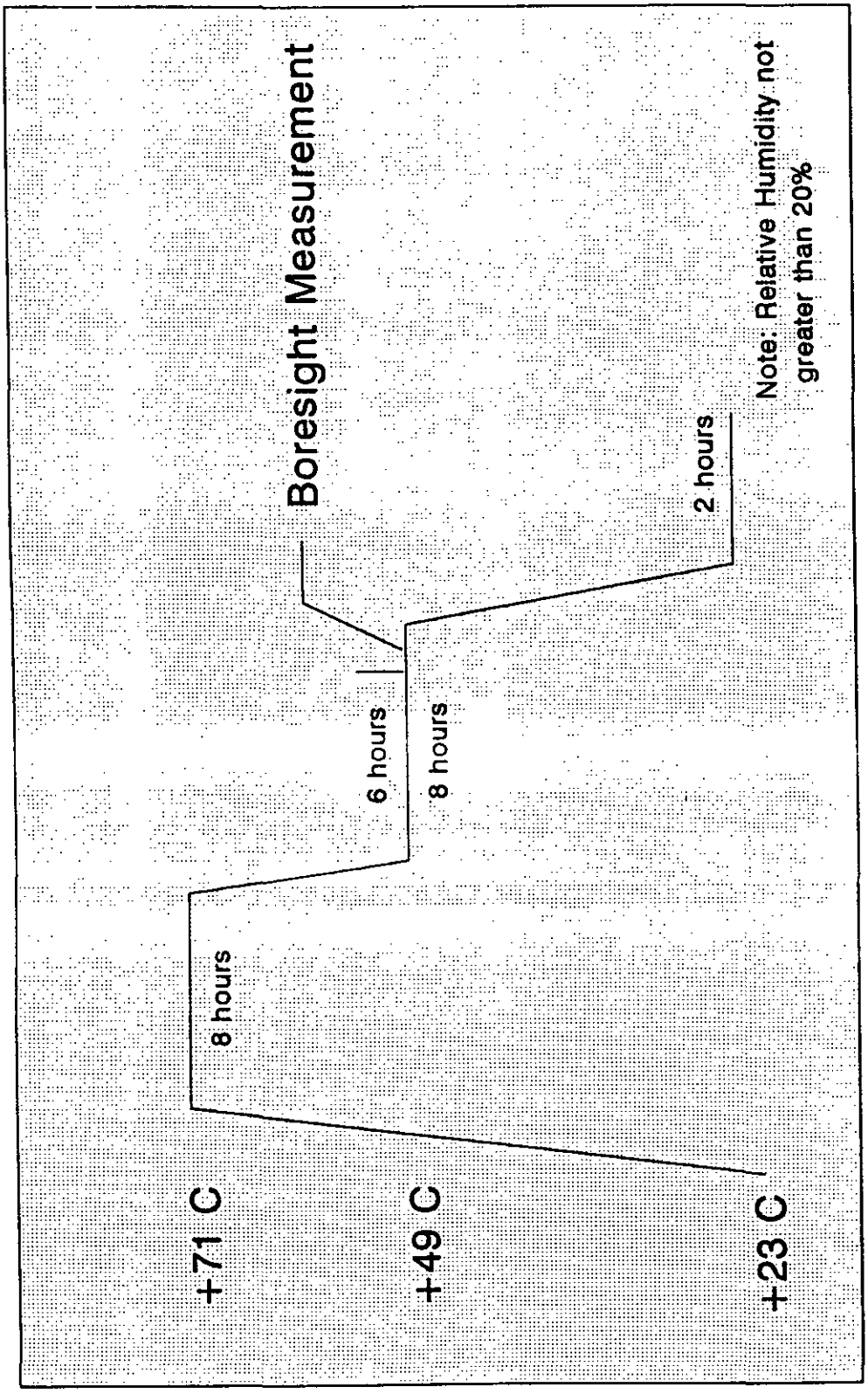



FIGURE 6
HIGH TEMPERATURE PROFILE



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-L-49320(CR) | 2. DOCUMENT DATE (YYMMDD)<br>14 April 1993 |
|---|---|---|

3. DOCUMENT TITLE: LASER INFRARED OBSERVATION SET, AN/PVS-6

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (*Last, First, Middle Initial*) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (*Include Zip Code*) | d. TELEPHONE (*Include Area Code*)<br>(1) Commercial<br>(2) AUTOVON (*If applicable*) | 7. DATE SUBMITTED (*YYMMDD*) |

8. PREPARING ACTIVITY

| a. NAME<br>U.S. ARMY CECOM<br>ATTN: AMSEL-LC-ED-TM | b. TELEPHONE (*Include Area Code*)<br>(1) Commercial<br>908-532-5851<br>(2) AUTOVON<br>992-5851 |
|---|---|
| c. ADDRESS (*Include Zip Code*)<br>Fort Monmouth, NJ 07703-5023 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |

DD Form 1426, OCT 89 Previous editions are obsolete 198-290

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